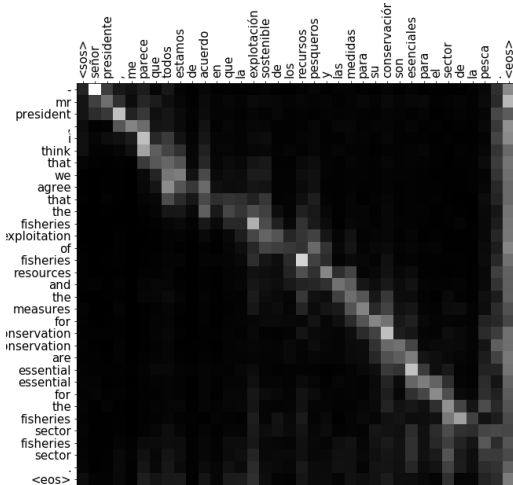


Natural Language Processing Tasks: Neural Machine Translation (NMT), Text-to-Code Generation, AI Agent, Fine-Tuning LLMs, Retrieval Augmented Generation (RAG), Sentiment Analysis, Text Extraction (OCR)

Neural Machine Translation

- Spanish → English Translation
- Developed an Encoder-Decoder architecture with Attention mechanism from scratch
- Trained and evaluated on parallel corpora, achieving a BLEU score of 25.37



Hierarchical Attention Network

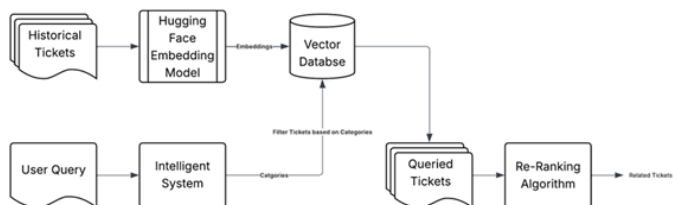
- Document Classification
- Implemented a custom Encoder architecture with hierarchical attention mechanisms
- Designed separate word-level and sentence-level attention layers from scratch

sentence_1 74milipeachbiden
sentence_2 **can't** how people do **not** want police until they need them
sentence_3 i **remember** them playing with the **hair** on my legs
sentence_4 back pedal file joe
sentence_5 joe
sentence_6 awful funny **news** will dub the climate as in crisis but will not address the boarder crisis as such

- Visualized attention weights to interpret model decisions (darker red = higher word importance, darker blue = higher sentence importance)

Ticket Checker

- Developed a deep learning-based multi-class ticket classification model
- Applied category-based filtering on a vector database using predicted labels
- Implemented a Retrieval-Augmented Generation (RAG) pipeline
- Generated ticket embeddings using HuggingFace SentenceTransformers
- Retrieved semantically similar tickets from a vector database
- Apply re-ranking algorithm on the retrived tickets
- Applied a re-ranking algorithm to improve retrieval relevance and response quality



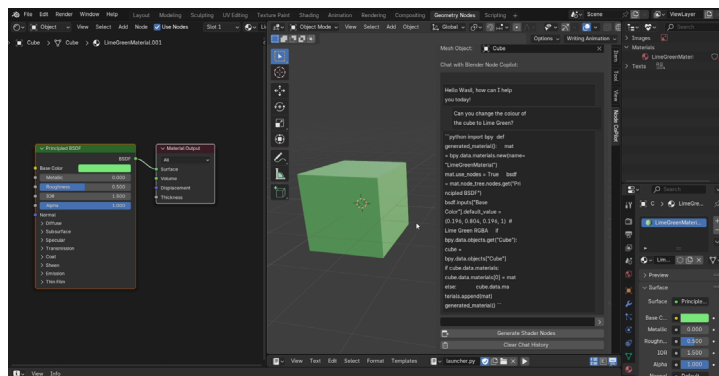
Theoretical Research on Search and Recommendation Systems

- Self-directed study of two-stage industry retrieval and ranking pipelines.
- Offline Retrieval:** Training Two-Tower Bi-Encoders via Contrastive Learning and Hard Negative Mining (clicked vs. non-clicked items).
- Indexing item embeddings in a Vector DB using ANN for scale.
- Pre-computing item embeddings and indexing them in a Vector Database using ANN (Approximate Nearest Neighbors) algorithms.
- Online Retrieval:** Embedding user queries into high-dimensional vectors through the trained Query Encoder.
- Executing a low-latency ANN search against the indexed item store to retrieve the Top-K most relevant candidates, effectively narrowing the search space for the high-precision ranking stage.
- Offline Ranking:** Architecting a Feature Store to manage User and Items.
- Training Neural Networks for CTR (Click-Through Rate) Prediction and intent detection via Pointwise/Pairwise loss.

- Online Ranking:** Dynamically augmenting the Top-K candidates with real-time features and historical state fetched from the Feature Store.
- Scoring enriched candidates via the Ranking Model and applying final sorting.

Blender Text-to-Code

- Developed a Text-to-Code ChatBot within Blender UI
- Translated natural language prompts into Geometry Nodes and Shader Node configurations
- Built an in-application chatbot interface inside Blender to enable real-time communication between the LLM and the 3D environment
- Integrated AWS-hosted foundation models for inference
- Fine-tuned a LLaMA-based model using Reinforcement Learning from Human Feedback (RLHF) with Proximal Policy Optimization (PPO)



Quality Agent

- AI-Powered Requirement Analysis System
- Developed a requirement analysis and validation tool using a multi-agent AI framework
- Designed specialized agents for: Automated test case generation, Requirement analysis and validation, Data extraction from Microsoft Word documents, Repository data retrieval via SVN API
- Integrated a self-service portal using JavaScript for user interaction and workflow automation

Vision + Language

- Built a multimodal VQA system integrating visual features with textual embeddings
- GloVe word embeddings for textual feature representation
- Compared LSTM, GRU, and Bidirectional architectures for sequence modeling

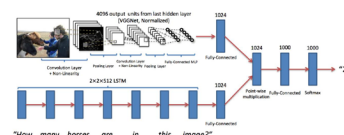


Table Extraction

- Extracted tables from PDFs using Amazon Textract and Azure Document Intelligence
- Converted table structures into Excel format for analysis
- Transformed tabular data into textual representations

Sentiment Analysis Projects

- Kaggle Competition: Quora Insincere Questions Classification and Bags of Words Meets Bags Of Popcorn.
- Utilized GloVe and Google Word2Vec embeddings for feature representation

Evaluation Metrics

- Accuracy
- BLEU Score
- F1 Score
- Area under the ROC curve (AUC)
- Qualitative Analysis